

ALI EZZAT

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WORK EXPERIENCE

Synapse Analytics

Principal Data Scientist

Cairo, Egypt
Mar 2019 - Present

- Founded and scaled the Data Science function from a single junior hire; grew and led a team of 10 data scientists (2 senior, 8 junior)
- Owned end-to-end modeling across multiple projects while mentoring junior data scientists and setting technical direction
- Delivered end-to-end ML systems across industrial manufacturing, supply chain, credit risk, and healthcare, driving significant cost reductions and automated decision-making for enterprise clients at scale
- Collaborated with data engineering to design data pipelines, docker-containerized model APIs and production deployment workflows
- Served as applied ML lead for Konan, the company's internal MLOps platform — translating real-world production workflows into platform requirements, supporting ~25 production models across ~15 client deployments

SELECTED PROJECTS

Credit Risk ML Platform — Multiple Financial Institutions

Machine learning-based credit scoring pipelines; processing 100K+ applications monthly, facilitated \$150M+ in loans to date; Reduced average default rates by 40% across all deployments

SMS Feature Extractor — Raya

Multilingual NLP pipeline that extracts financial signals from unstructured SMS data for alternative credit scoring of thin-file applicants

Collection Bot — Forsa

Automated WhatsApp conversation engine (via Twilio + GPT-4o) for loan applications; contacts overdue applicants, updates contact details, and drives repayment

Industrial Optimization — Unilever

Real-time recommendation system operating on live sensor data from a detergent-manufacturing tower; Achieved a 23% reduction in monthly natural gas costs

Supply Network Optimization — Procter & Gamble

Supply network design (SND) optimization tool that recommends ideal warehouse and hub placements to minimize total logistics costs across a given store network; Achieved a 18% reduction in total supply network cost versus the client's existing configuration

Withdrawal Amount Prediction — Western Union

Time-series forecasting model for daily cash withdrawal volumes per branch; Reduced over-supply error rate from 300% to 20%

Magnet Point Optimization — London Cab

Performed temporal clustering to determine magnet points (parking spots) that minimize time taken by drivers to arrive at pickup locations; Reduced overall drivers' pickup times by 13%

SKILLS

Programming:	Python, R, SQL, MATLAB
Libraries:	PyTorch, scikit-learn, XGBoost, pandas, NumPy, Plotly, tidyverse, dplyr, ggplot2, Shiny
AI:	NLP, LLMs, RAG, Conversational AI, QLoRA fine-tuning, Transformers, LangChain
Competencies:	Optimization, Recommender Systems, Causal Inference, Time-Series Forecasting
DevOps:	AWS, Docker, Git, Jupyter, VS Code, Neo4j

EDUCATION

Nanyang Technological University

PhD Computer Science

Singapore
Aug 2013 - Jun 2018

Nanyang Technological University

MSc Bioinformatics

Singapore
Aug 2011 - Dec 2012

Ain Shams University

BSc Computer and Information Sciences

Cairo, Egypt
Sep 2002 - Aug 2006